

Identifying the boundary between dialects and languages: The role of similarity and usage

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ARTICLE INFO

Keywords:

Dialects
Bidialectalism
Bilingualism
Lexical selection
Picture-word interference
Lexical organization

ABSTRACT

The question of how to distinguish between dialects and languages has plagued linguists for decades. The present investigation uses behavioral observations to objectively distinguish between bilingual and bidialectal translation effects. Using the picture-word interference paradigm, Italian – Neapolitan (Exp 1) bidialectal participants named pictures while ignoring semantically identical, related, or unrelated distractor words presented in their two linguistic varieties. Consistent with prior investigations of dialectal lexical selection studies, we observed translation equivalent interference between dialectal varieties that were mutually non-intelligible and in participants who were active, proficient speakers of both varieties. Experiment 2 assessed the distractor effects in Italian-English bilingual speakers and did not observe similar translation equivalent inference effects. The results suggest that mutual intelligibility and high speaker proficiency are not necessary and sufficient conditions for establishing independent lexical systems.

1. Introduction

In bilingualism research, the definition of a bilingual can range from someone raised in a bilingual environment, with high proficiency in both languages, or it can include someone in the process of learning a 2nd language, and many points in between. Despite the inclusive nature of the term bilingual, definitions typically do not include those individuals who speak two variants of a single language, the so-called *bidialectal*. This is despite the fact that the distinction between languages and dialects is extremely muddy and there may be examples of bidialectal speakers who are, for all intents and purposes, bilingual. Indeed, there is a small but growing body of research aimed to understand whether bidialectal speakers are effectively bilingual, from the perspective of language representation and processing. Bidialectal speakers, like bilingual speakers, choose between distinct words, grammatical structures, and phonological alternatives. How they achieve this feat is unclear and may well depend on a number of factors, such as the degree of similarity between their two variants, a speaker's usage patterns for the two variants, etc. The aim of the present research is to explore two factors proposed to influence how linguistic variants are represented and processed, to better understand how to objectively discriminate between dialectal varieties and language varieties, from a representational and processing perspective.

1.1. Distinguishing between dialects and languages

Dialects are sometimes defined as a *regional variety of a language* (e.g., Merriam-Webster dictionary), but such a definition does nothing to help identify the boundary between dialects and languages. Indeed, one would not want to define French as a regional variety of Italian, but many are content to identify Scots as a regional variety of English. What is the crucial difference between these two pairs of variants? Over the years, linguists have referred to criteria such as prestige, size, and mutual intelligibility as crucial features that distinguish languages from dialects, but in practice these factors fail to provide a satisfactory distinction (Hudson, 1996; Wei, 2000). For example, Danish, Norwegian and Swedish are classified as distinct languages, ostensibly because they are of equal prestige in their respective countries, despite having high levels of mutual intelligibility (Gooskens & van Heuven, 2021). But, are psychological processes influenced by politically-determined geographical boundaries?

Some prior research has examined whether particular pairs of linguistic varieties were represented and processed like separate languages or like varieties of the same language by assessing performance on well-established psychological paradigms (Chen & Zhou, 2022; Kirk, Kempe, Scott-Brown, Philipp, & Declerck, 2018; Melinger, 2018, 2021; Woutersen, Cox, Weltens, & De Bot, 1994). Repetition priming describes the

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<https://doi.org/10.1016/j.cognition.2025.106388>

Received 16 May 2025; Received in revised form 29 November 2025; Accepted 3 December 2025

Available online 15 December 2025

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observation that a second retrieval of a representation is achieved faster or more accurately than the first retrieval. These effects can persist despite formal changes between the first and second presentations, e.g., both spoken and written forms can be used between presentations (see Henson, Eckstein, Waszak, Frings, & Horner, 2014 for a review). However, repetition priming does not emerge between languages (Monsell, Matthews, & Miller, 1992), suggesting that translation equivalents do not share a common lexical representation. Woutersen et al. investigated inter-dialectal repetition priming effects in speakers of the Maastricht dialect of Dutch with the aim of determining whether they were processing their two varieties as separate languages. They observed that active users of the Maastricht dialect did not show any repetition priming between dialectal and standard Dutch words while passive users of the dialect, namely speakers who had exposure to the Maastricht dialect but did not actively speak the dialect, did produce repetition priming effects. This led Woutersen et al. to propose that the active dialect speakers had established bilingual-like representations for their two varieties of Dutch while the passive users had not.

Woutersen et al.'s (1994) study established a methodology for objectively identifying bilingual vs. dialectal processing characteristics. This same approach was extended to the language production context by Melinger (2018, see also Kirk et al., 2018 for an example of this logic applied to the language switching paradigm). Melinger used the picture-word interference (PWI) task to investigate lexical selection processes in bidialectal speakers. In this task, participants are asked to name a picture while ignoring a visually or auditorily presented distractor word. The relationship between the distractor word and the picture can be systematically manipulated, giving this paradigm great flexibility and investigative power. Commonly, distractor words can be categorically related to the target picture (e.g., picture = TABLE, distractor = chair), phonologically related to the target picture (e.g., picture = TABLE, distractor = tailor), or unrelated (e.g., picture = TABLE, distractor = choir). Picture naming times are slower when the distractor word is categorically related to the target compared to the unrelated condition while they are faster when the distractor word is phonologically related (Glaser & Döngelhoff, 1984; Schriefers, Meyer, & Levelt, 1990). In a bilingual variant of this task, bilingual speakers are asked to name a picture in one of their languages while ignoring a distractor word displaying the name of the picture in their other language (e.g., picture = TABLE, distractor = mesa (the Spanish word for table) displayed to English-Spanish bilinguals). Here, bilingual speakers exhibit a *translation equivalent facilitation effect* as their picture naming times are faster compared to an unrelated distractor condition (Costa & Caramazza, 1999; Costa, Miozzo, & Caramazza, 1999, see Hall, 2011 for a review). This important and surprising result has been replicated with highly related languages (e.g., Spanish – Catalan) as well as more dissimilar languages (e.g., Spanish – English, Swedish – English). It has also been observed for early bilinguals living in diglossic contexts (e.g., Spanish – Catalan) and for late learners living in a single language environment (Dylman & Barry, 2018; Hermans, 2004; Roelofs, Piai, & Rodríguez, 2011; Roelofs, Piai, Rodríguez, & Chwilla, 2016). Dylman and Barry report larger translation equivalent facilitation effects when participants name pictures in their L2 compared to their L1 (see also Costa & Caramazza, 1999). Translation equivalent facilitation has been observed when the experiment is conducted in pure or mixed blocks (Costa et al., 1999; Tomoschuk, Ferreira, & Gollan, 2021), with distractors only from the non-target language (Roelofs et al., 2016) and with and without the inclusion of the semantically related condition (Dylman & Barry, 2018). Recently, Feleke (2024) observed translation equivalent facilitation between Oromo and Amharic, two languages of Ethiopia, using this paradigm. Finally, the translation equivalent facilitation effect has been observed both at negative SOAs and with simultaneous presentation of the picture and written distractor (Costa et al., 1999).

Melinger (2018) examined this translation equivalent distractor effect in a bidialectal context. Specifically, she asked bidialectal Standard Scottish English (SSE) - Scots dialect speakers to name pictures in their

preferred dialect while ignoring the non-cognate translation equivalent picture labels (e.g., picture = TROUSERS, distractor = breeks (Scots word for trousers)) presented auditorily. Across multiple experiments, she reported significant *dialect equivalent interference*, suggesting that the bidialectal participants in her experiments used a different mechanism to select between dialectal alternatives than bilingual speakers use to select between translation equivalents.

The dialect equivalent interference effect appears to be robust, having been replicated with other English varieties (specifically British vs. American dialects), with registers (e.g., formal vs. informal picture labels), and persists despite some methodological changes (Melinger, 2021). Specifically, Melinger (2018) used auditory distractors, because the dialects she investigated did not have a standard written form. Melinger (2021) then explored the contribution of the distractor modality, as all previous demonstrations of the translation equivalent facilitation effect had been obtained with written distractors. Across three experiments using the same stimuli, Melinger compared dialect equivalent translation effects when the distractor words were spoken in a Scottish accent, spoken in an American accent, or written. This comparison was possible because both British and American variants of English have established orthographies, unlike the Scottish dialects she investigated previously. Here, she again observed dialect equivalent translation interference for both Scottish and American spoken distractors, although the latter induced overall slower naming times. Critically, written distractors also induced interference, although the effect was smaller and less robust than with spoken distractors. Consequently, Melinger concluded that the polarity reversal observed for languages compared to dialects could not be explained by the modality of the distractor words.

However, Melinger (2018) also acknowledged that dialects of English may not represent the strongest test case for identifying language-like facilitation effects. She noted that, despite using distinct, non-cognate lexical alternatives in her studies, the system-wide differences between the tested English varieties may not have been sufficient to warrant the establishment of separate independent linguistic systems. She also acknowledged that characteristics of the experimental context, specifically testing in a formal laboratory setting, recruiting undergraduate participants who may inhibit dialect usage in the university setting, may also have contributed to the emergence of an interference, rather than a facilitation, effect. To exclude these possibilities, she suggested testing other populations of dialect users, particularly high proficiency, active dialect speakers of varieties with lower mutual intelligibility, outside the laboratory.

1.2. Mutual intelligibility and formal similarity

Mutual intelligibility describes the extent to which speakers of different varieties can understand each other. Language similarity is a measure of the formal overlap between two varieties (Gooskens & van Heuven, 2021). These two measures are not equivalent, as formally similar varieties may still be unintelligible, for example if similarities are masked by phonological alternations. The extent of exposure to varieties may also be unequal, rendering one variant as more familiar, leading to unequal contact-based intelligibility. The degree of mutual intelligibility between two varieties influences their classification as dialects or languages, but it fails to provide a satisfactory classification by itself. Nevertheless, it may be a factor that contributes to how two varieties are mentally represented and processed by speakers, especially as mutual intelligibility implies a certain amount of formal overlap. If two varieties are sufficiently similar as to be reciprocally understood, the impetus to establish independent phonological, lexical, or grammatical representations may be diminished. Similarly, if the divergences between two varieties follow regular patterns, accommodation processes may suffice to process both without establishing distinct systems.

The degree of similarity between two languages can impact on language processes in myriad of ways. For instance, similarities between an

L1 and an L2 can influence learning outcomes and learning strategies (Marian, Bartolotti, van den Berg and Hayakawa, 2021). Language learning is facilitated by language similarity – learning an L2 that is lexically and structurally similar to the L1 is easier than learning a dissimilar L2 (U.S. Department of State, Foreign Service Institute, n.d.; Hayakawa, Bartolotti, & Marian, 2021). This may be because related languages do not need to be represented totally independently. Likewise, some studies suggest that similar representations may be shared across languages (e.g., cognates; De Bot, 1992; structural alternatives; Hartsuiker, Pickering, & Veltkamp, 2004). In an investigation of how similarity affects representational autonomy between languages, Huang et al. (2019) found that the magnitude of structural priming effects was unaffected by language similarity when the described events were unrelated, suggesting that similarity did not influence the degree of representational independence. However, when the prime and target involved the same event (e.g., both involved *giving* events), priming was greater for related languages, suggesting that similarity did influence the integration between lexical and syntactic information.

There is no question that similarity between words in two languages influences processing (cf. cognate facilitation, (Costa, Caramazza, & Sebastian-Galles, 2000), false friends (Lemhöfer & Dijkstra, 2004), etc.). Furthermore, the degree of similarity between individual words can influence the magnitude of facilitation effects. For example, identical cognates produce larger facilitation effects than non-identical cognates in word recognition (Dijkstra & Van Heuven, 2002a; Dijkstra & van Heuven, 2002b).

On the other hand, many experimental effects have been obtained with both similar and dissimilar languages, with comparable magnitudes (see Kantola & van Gompel, 2011, Schoonbaert, Hartsuiker, & Pickering, 2007 for examples from the syntactic priming literature). Indeed, some effects, such as lexical competition in spoken word recognition, have even been observed between languages that do not share a modality (Shook & Marian, 2012). In most PWI studies showing between-language translation equivalent facilitation, related language pairs were used (e.g., English - Spanish, Swedish - English, Dutch - English, Spanish - Catalan; Costa et al., 1999; Costa & Caramazza, 1999; Dylman & Barry, 2018; Hermans, 2004, but see Feleke, 2024 and Emmorey, Mott, Meade, Holcomb, & Midgley, 2021 for evidence of translation equivalent facilitation from unrelated languages, namely Oromo and Amharic in the former and ALS-English in the latter), although no direct comparison of the magnitude of the effects were made.

Two related varieties that have a high rate of formal overlap in lexical and/or structural representations may end up with a majority of their representations being shared, leading to a co-dependent lexical organization, as proposed by Labov (1998). This continuum of shared-distinct representations may be a factor as to whether two varieties are then process as two variants of a single language or as two distinct languages.

1.3. Proficiency and active vs. passive usage

Another factor that could influence the representational status of dialectal alternatives is whether or not a speaker actively uses the dialect or is only passively familiar with it.

Sumner and Samuel (2009) investigated the representations of r-less dialectal forms of the New York City dialect (e.g., baker – bak[ə]). They distinguished their participants as Overt or Covert speakers. Overt speakers produced r-less forms in their own extemporaneous speech while Covert speakers produced r-full utterances, despite both groups being born and raised in the New York area. In a series of repetition and semantic priming experiments, Sumner and Samuel found contrastive patterns of behavior for Overt and Covert dialect users, arguing that only the Overt dialect speakers established independent access representations for the r-less and r-full variants; Covert dialect users established a single r-full access representation and mapped both variants to that

single representation.

In the production literature, Kirk et al. (2018) investigated the control mechanism that bidialectal speakers utilize to select one lexica over the other in a language switching task. They categorized their bidialectal participants as *active* or *passive* dialect users, based on a post-test rating of the ‘nativeness’ of the dialectal utterances produced during the experiment. They also included a monodialectal group who were trained in the stimuli prior to testing. Both dialectal groups produced symmetrical switching costs while the monodialectal group produced asymmetrical switch costs, suggesting usage patterns were not driving lexical organization.

In Melinger’s (2018) PWI studies, participants were not categorized as active or passive users but many participants did report that they rarely spoke Scots themselves and many expressed negative views of Scots, characterizing it as slang rather than a dialect or language. For investigations of British vs American varieties of English (Melinger, 2021), none of the participants were active users of the American dialect but had passive exposure through literature, music and other media. Therefore, based on these self-reports, it seems that the participants in these studies were unlikely to be highly proficiency active users of both tested varieties. This may explain why these participants did not show evidence of having established language-like selection processes. Indeed, Geukes and Zwitserlood (2016) demonstrated that newly learned words from an artificial language also produce translation equivalent interference, suggesting that low proficiency or limited user experience with a language could be associated with interference in these experimental contexts. Broersma, Carter, and Acheson (2016) also found that proficiency modulated the polarity of cognate effects in picture naming, with higher proficiency leading to a facilitation effect while lower proficiency led to an inhibition effect as the similarity between words increased. This finding also implicates the potential importance of proficiency to bilingual lexical selection.

Active vs. passive usage may be correlated with language proficiency, as one builds proficiency not only through exposure but also through use. Proficiency in an L2 has implications for processing and behavior. For example, it has been proposed that the activation within a linguistic system is linked to a speaker’s proficiency. In turn, the strength of inhibition needed to switch out of that language will also be determined by proficiency (e.g. Costa & Santesteban, 2004; Meuter & Allport, 1999, however see Blanco-Elorrieta & Caramazza, 2021 for an alternative perspective). Hence, it is important to consider both proficiency levels as well as usage patterns.

1.4. Present study

In the current study, we continue the research programme initiated by Melinger (2018) which asked whether bidialectal speakers exhibit the same pattern of behavioral effects as bilingual speakers. We advance this research programme by examining linguistic variants that represent features typically associated with languages. Specifically, we focus on speakers who are proficient, active speakers of both of their varieties and on linguistic varieties with low levels of mutual intelligibility. We use self-report measures of proficiency and usage rate word familiarity scores. Gooskens and Swarte (2017) report that self-report measures are highly correlated with performance measures of proficiency (but see Tomoschuk, Ferreira, and Gollan (2019) for arguments against the reliance on self-report measures).

Here, speakers of Italian and Neapolitan varieties were tested, using the same PWI design originally developed by Costa et al. (1999) and subsequently used by Melinger (2018, 2021). Neapolitan is a variety of Italian spoken in Naples and the surrounding areas. It has marked dissimilarities to the standard variety, including distinct grammatical and morphosyntactic rules and contrasting phonology that masks lexical similarities. According to the Endangered Languages Alliance (ELAlliance, n.d.), standard Italian and Neapolitan are not mutually intelligible. In contrast to the English varieties previously investigated,

Neapolitan has relatively high prestige, and is used across contexts (Crocco, Gili Fivela, & D'Imperio, 2022; Dal Negro, 2008). However, like many non-standard varieties, the linguistic status of Neapolitan is debated in the literature. Some researchers classify Neapolitan as a dialect of Italian rather than a distinct language (e.g., Boga, 2020) on the basis of relatively close phonemic similarities. Boga (2020) assessed 58 Romance varieties, with a focus on varieties in Italy, clustering these varieties into language-language pairs, language-dialect pairs, or dialect-dialect pairs. The latter captures two varieties of the same language family. Distance and similarity scores for a core set of 110 common concepts were calculated using the Levenshtein (Levenshtein, 1966) and Needleman-Wunsch (Needleman & Wunsch, 1970) methods. Plotting these distances onto a multi-dimensional space, Boga's results placed Neapolitan and Standard Italian as spatially close, in the cluster with the highest similarity and lowest distance.

Taking a more social-psychological approach to the question of how to distinguish languages and dialects, Kumove (2020) examined the relationship between social trust and communication potential with the latter being linked to mutual intelligibility. In their study, participants completed a language 'barometer', which included a question about native language. Kumove reported that no Italian respondent reported speaking a regional variety such as Neapolitan, reflecting the Italian practice of considering regional varieties as dialects while elevating standard Italian to the status of language. If the psychological processes of linguistic varieties are sensitive to these cultural or attitudinal factors, then Neapolitan would be processed as a dialect of Standard Italian.

In contrast to the above, other researchers have argued that the historical roots of Neapolitan, which run in parallel to modern Italian, render it a language in its own right (Feliu, 2017; Ryan, 2018). Following the logic outlined in Melinger (2018), experimental evidence can be brought to bear on this open question, providing objective evidence one way or the other.

Given the ongoing discussion regarding the relationship between Neapolitan and Italian, as well as the relatively high status, wide-ranging usage patterns in the community, and reported low mutual intelligibility between the variants, this pair was selected as a promising next step for the investigation of dialectal processing. Following Sumner and Samuel (2009), we might observe that dialect equivalent distractor effects shift towards facilitation when dialect usage rates are high, as in this sample, resulting in more 'language-like' distractor effects compared to what has been observed previously (Melinger, 2018, 2021). The results from Geukes & Zwitterlood, 2016 also suggest the higher proficiency levels generally associated with this sample could impact the polarity of distractor effects. Such a result would support the claim that active, proficient, usage is required to set up independent lexical organizations between linguistic varieties. The explicit premise of this investigation is that translation equivalent facilitation signals bilingual processing and lexical organization while dialect equivalent interference signals monolingual, dialectal, processing (but see discussion of Dylman and Barry (2018) below for a refinement of this proposition). Therefore, if dialect equivalent distractor interference is observed despite high usage and proficiency scores, then this would call into question the role of usage and proficiency in distinguishing dialects from languages. Specifically, it would suggest that, for this specific variety, active use and high proficiency are insufficient to support the establishment of independent lexical systems, relying instead on the co-dependence that Labov (1998) and subsequently Melinger (2018) argued is a hallmark of dialectal linguistic organization.

Similarly, if increased formal distance between varieties contributes to the need for independent lexical systems, then we might expect Italian-Neapolitan speaker to produce 'language-like' translation equivalent facilitation effects, supporting the idea that Neapolitan is represented and processed like a distinct language. In contrast, if formal properties are insufficient to support independent representation when other language-like properties are still absent, e.g., cultural and organizational support, then we might again expect to find translation/

dialect equivalent interference, suggesting that Neapolitan is represented and processed like a dialect of Italian.

2. Experiment 1: Italian vs. Neapolitan

Experiment 1 investigates translation equivalent distractor effects in a new linguistic context and population. Italian participants who are proficient speakers of Neapolitan were tested, using the design and procedure from Melinger (2018). Participants were recruited from the local community rather than from an undergraduate psychology subject pool and they were tested in familiar settings, such as their own homes, by a native speaker of Italian. Dialect usage in this community was predicted to be higher than in the samples used previously. Furthermore, as Italian and Neapolitan are not mutually intelligible yet both have relatively high prestige (Crocco et al., 2022; Dal Negro, 2008), our particular interest will be to determine whether the polarity or magnitude of the translation equivalent distractor effects are dialect-like, i.e., slowing naming times, or language-like, i.e., facilitating naming times, in this group.

2.1. Methods

2.1.1. Participants

Twenty-five Italian-Neapolitan bi-dialectal speakers (17 females and 8 males) aged between 18 and 60 (mean age = 35) from Agerola, Italy, which is near the Amalfi coast and falls within the wider Neapolitan metropolitan area, were recruited by word of mouth to take part in this study. All participants were monolingual native Italian speakers who grew up in Italy and were familiar with the local Neapolitan variant. The sample size was set without a power analysis, based on the typical number of participants tested for these types of studies. A subsequent sensitivity power analysis was conducted using the Pangea web-based power analysis tool for repeated measures designs (jakewestfall.shinyapps.io/pangea/). We assessed the minimum effect size for the interaction between dialect and distractor relation we could achieve 80 % power for given our existing sample size. Using the default variance estimates provided, the sensitivity analysis showed we could achieve 81.4 % power for an effect size (d) of 0.36 with 25 subjects and 32 items, each presented twice in every condition (Westfall, Kenny, & Judd, 2014). For paired comparisons (e.g., to detect translation equivalent facilitation or interference) we could achieve 80 % power to detect an effect size of 0.53.

The language background questionnaire, described below, revealed that 8 of our participants reported speaking Neapolitan more often than standard Italian, 10 spoke Standard Italian more than Neapolitan, and 6 spoke the two variants with equal frequency (one participant did not respond). Participants were asked to self-rate their proficiency both for speaking and listening to Standard Italian and Neapolitan, on a 5-point scale. The average self-rated spoken Neapolitan proficiency was 3.6 (range from 1 to 5, $SD = 1.4$) while the average self-rated spoken Standard Italian proficiency was 3.8 (range from 2 to 5, $SD = 0.9$). The mean difference in the two proficiency ratings is 0.2. Seven participants rated themselves more proficient when speaking Neapolitan, 9 rated their proficiency as equally good in the two variants, and 9 rated their Standard Italian proficiency as higher than their Neapolitan proficiency. Taken together, these data from our language background questionnaire support the premise that this sample reflects active, proficient users of Neapolitan, although proficiency and usage rates did vary. The data from one participant was excluded from the final analysis due to a technical error that led to the data being incomplete. The experimental procedure was approved by the University of Dundee Psychology Ethics Committee and participant gave written informed consent prior to testing.

2.1.2. Materials

Thirty-two photographs of common objects drawn from 15

categories were selected, see Fig. 1 for examples. All objects had phonologically distinct Italian and Neapolitan labels. A Neapolitan consultant oversaw materials selection.

Each picture was paired with six auditorily presented distractor words. Distractor words were either Italian words that corresponded to the picture labels (e.g., Carota ‘carrot’) or their Neapolitan translation equivalents (e.g., Pastenacchia ‘carrot’). Distractor words either referred to the same meaning as the target picture (e.g., target = CARROT, distractors = carota OR pastenacchia), a semantically related meaning (e.g., target = CARROT, distractors = sedano OR laccia ‘celery’), or an unrelated meaning (e.g., target = CARROT, distractors = strega or janara ‘witch’). Spurious semantic, associative, phonological or orthographic relationship between the distractor words and the target picture names were avoided. Because the Neapolitan picture labels were never produced in this experiment, phonological overlap between the Neapolitan picture name and distractor words was allowed. A full set of stimuli can be found in Appendix A.

Using Adobe Photoshop, pictures were scaled to fit within a 3.5 cm square and always appeared centered on the computer screen. Distractors were presented auditorily over headphones at an SOA of -150 ms (Damian & Martin, 1999). Distractor words were recorded by an Italian male with a noticeable Neapolitan accent. Each individual sound file was edited to include 200 ms of silence before each word onset and a cue was embedded in the sound file at 350 ms (150 ms post-speech onset), which triggered the presentation of the target picture.

Stimuli were presented in 2 blocks consisting of 192 trials each, with the order of trials in each block varying across participants. Each of the 32 pictures appeared in each of the six conditions, in each of the two blocks, for a total of 12 repetitions of each picture across the experiment. The block trials were pseudo-randomized with the restriction that the same picture, the same distractor word, the beginning phoneme of picture names, and the picture category did not appear in consecutive trials. There was also the restriction that stimuli from the same experimental condition could appear in no more than two consecutive trials. A further control was that the distractor word in trial n could not be the name of the picture in trial $n + 1$.

Following the main body of the experiment, participants completed a language usage questionnaire and a word knowledge test, assessing familiarity with the Neapolitan vocabulary used in the experiment. The former was adapted from the questionnaire used in Melinger (2018) and asked participants to self-rate their proficiency in speaking and listening to both Italian and Neapolitan. It also asked about dialect usage at home during childhood, current frequency of use of their two varieties, and ease of understanding people from different regions of Italy. The word knowledge test presented participants with the Neapolitan picture labels from the experiment and ask them to select one answer from a choice of a) I know and use this word, b) I know but don't use this word, or c) I don't know this word.

2.1.3. Procedure

The experimental procedure also replicated that of Melinger (2018). Specifically, trials consisted of a central fixation cross (500 ms), followed by a blank screen interval (250 ms), after which the distractor word was presented over headphones. 150 ms after the onset of the distractor word, the target picture appeared for a naming response. The picture was presented for 2000 ms. Participants were instructed to name the picture as quickly and accurately as possible while ignoring the distractor words. Naming latencies were measured with a voice key and each response was recorded into an individual .wav file. These .wav files were subsequently inspected for accuracy using CheckVocal (Protopapas, 2007). The entire testing session lasted approximately 45 min. Presentation of the experiment was controlled using DMDX software. Unlike in Melinger (2018), participants in this study were tested in their homes, rather than in a laboratory.

Prior to the main experimental blocks, participants were familiarized with the materials by showing them a booklet containing the

experimental images and their Italian labels we asked them to use in the study. Participants could look at this booklet for as long as they wanted. Following this familiarization phase, participants practiced naming each picture once. If participants produced any errors during the practice block, the experimenter used the printed booklet to remind them of the correct Italian picture labels. Then, 16 practice trials which included the presentation of a distractor words from the unrelated conditions were presented. No overt mention of Neapolitan was made at any point in the instructions to the experiment. Following the completion of the main experiment, participants completed the language usage questionnaire and word knowledge test with pen and paper. A total of 384 trials were recorded for each participant, with a total of 9600 trials in total.

2.2. Results

Table 1 reports the average rating for the frequency of use of each variant as well as the self-assessed proficiency in listening and speaking both Italian and Neapolitan. Paired t -tests reveal that this sample report using both variants with equal frequency, $t(23) = 0.43$, $p = .671$, and their self-assessed proficiency in each variety was also comparable, $t(24) = 0.6$, $p = .553$. On average, participants reported being familiar with 91.6 % of the dialect words used in the experiment.

Mean response times (RTs) for correct trials, standard deviations, and mean proportion of errors in the experimental conditions are presented in Table 2, with the data visualized in Fig. 2. Trials in which the participant produced the wrong word, stuttered, hesitated, or failed to respond, and trials with voice key failures, including any trial with RT faster than 250 ms, were classified as errors and discarded from the RT analysis ($N = 606$; 6.3 %). Additionally, any trial which was 2.5 standard deviations from a participant's condition mean was treated as an outlier ($N = 210$; 2.2 %). In total, 8784 trials were included in the final analysis. As error rates were low (6 %), they were not analysed further.

Linear mixed effects models were used to analyse the reaction times across conditions. Analyses were conducted in JASP version 0.18.1 and p -values were approximated using the Satterthwaite method. The model included *Dialect* (Italian, Neapolitan) and *Relatedness* (same meaning, related meaning, unrelated) as categorical fixed factors. Participants and Items were included as random effects. The final model included random slopes of *Relatedness* by subjects.¹ Follow-up comparisons on the model's estimated marginal means used the Satterthwaite approximation for degrees of freedom, and the Tukey correction for multiple comparisons.

The results, presented in Table 3, revealed a main effect of *Relatedness* which is driven both by the comparison of the same meaning conditions compared to the unrelated conditions and the related meaning conditions compared to the unrelated conditions. Estimated marginal means (averaged across *Dialect*) revealed that participants named pictures slower in the context of related meaning distractors relative to the unrelated conditions, $M_{diff} = 36$ ms, $z = 8.3$, $SE = 4.3$, $p < .001$, while there was no reliable difference in naming times in the same meaning conditions relative to the unrelated condition, $M_{diff} = 6$ ms, $z < 1$, $SE = 9.0$, $p = .78$.

The main effect of *Dialect* was also significant. Estimated marginal means (averaged across *Relatedness*) confirmed that picture naming times were overall faster when the distractor was presented in Italian compared to Neapolitan, $M_{diff} = 47$ ms, $z = 15.0$, $SE = 3.2$, $p < .001$.

The interactions between *Relatedness* and *Dialect* were significant both when comparing the same meaning conditions to the unrelated conditions and when comparing the related meaning conditions to the unrelated conditions. These interactions are calculated on the basis of the omnibus model fit and therefore do not necessarily reflect the respective 2×2 interactions in isolation. To clarify these interactions, two additional LMEs were conducted with a reduced fixed effects

¹ $RT \sim Related * Dialect + (1 + Related | subject) + (1 | picture)$



Fig. 1. Example images which were presented with auditory distractor words.

Table 1
Mean (SD) rating estimates for the frequency of use of each variant, the self-reported proficiency in each variant, and the mean number of experimental dialect words that participants identified as familiar.

	Italian	Neapolitan
Frequency of use	4.0 (0.8)	4.0 (1.0)
Listening Proficiency	4.0 (0.7)	3.9 (1.1)
Speaking Proficiency	3.8 (0.8)	3.7 (1.3)
Dialect word familiarity		91.6 % (9.7)

Table 2
Mean response times (RTs) for correct trials, standard deviations and mean proportion of errors in the experimental conditions.

Distractor Conditions	Dialect			
	Italian		Neapolitan	
	RT (errs)	SD	RT (errs)	SD
Same meaning	891 (0.03)	107	990 (0.07)	91
Related meaning	970 (0.07)	83	996 (0.08)	86
Unrelated	939 (0.06)	87	955 (0.06)	83
Identity effect (unrelated – same meaning)	+48 (0.03)		–35 (–0.01)	
Categorical Interference (unrelated – related meaning)	–31 (–0.01)		–41 (–0.02)	

structure. When the related meaning conditions were removed from the analysis, the *Relatedness* $\times$ *Dialect* interaction remained significant, $t(5812) = 10.8$, $B = 20.5$, $SE = 1.9$, $p < .01$, suggesting that the influence of the same meaning distractor is modulated by Dialect. When the same meaning condition was removed, the *Relatedness* $\times$ *Dialect* interaction

was not significant, $t(5721) = 1.2$, $B = 2.3$, $SE = 2.0$, $p = .247$, reflecting the fact that the size of the semantic interference effect is comparable across dialects.

Planned comparisons investigated the nature of the *Dialect* $\times$ *Relatedness* interactions. Italian bidialectal speakers named pictures more slowly when the Neapolitan distractor word conveyed the same meaning as the target picture, relative to the Neapolitan unrelated condition, $M_{diff} = 35.4$ ms, $z = 3.6$, $SE = 9.9$, $CI [16.0, 54.7]$, $p < .001$, while the Italian same meaning distractors sped picture naming times relative to the Italian unrelated distractors, $M_{diff} = -47$ ms, $z = -4.7$, $SE = 9.9$, $CI [-66.1, -27.5]$, $p = .001$. Additionally, both Italian and Neapolitan related distractors slowed picture naming times relative to their dialect-specific controls. Traditional semantic interference effects were observed Italian distractors, $M_{diff} = 31.1$, $z = 5.3$, $SE = 5.8$, $CI [19, 43.0]$, $p < .001$, and from Neapolitan distractors, $M_{diff} = 40.1$, $z = 6.9$, $SE = 5.8$, $CI [28.6, 51.5]$, $p < .001$.

2.3. Discussion

As reported for previously studied dialect pairs, these results reveal dialect equivalent interference. The results are distinct from the language equivalent translation facilitation effects repeatedly observed in bilingual experiments. Hence, these findings suggest that Neapolitan is represented and processed like a dialect of Italian rather than a language distinct from Italian. This experiment represents an important observation of translation equivalent dialect interference between non-English linguistic varieties. This is an important novel observation, because Neapolitan was selected for this investigation because of its wide usage, relatively high prestige, and low mutual intelligibility with standard Italian. Despite these characteristics, it appears that the speakers tested here did not represent their two linguistic varieties as distinct languages but rather have established a co-dependent

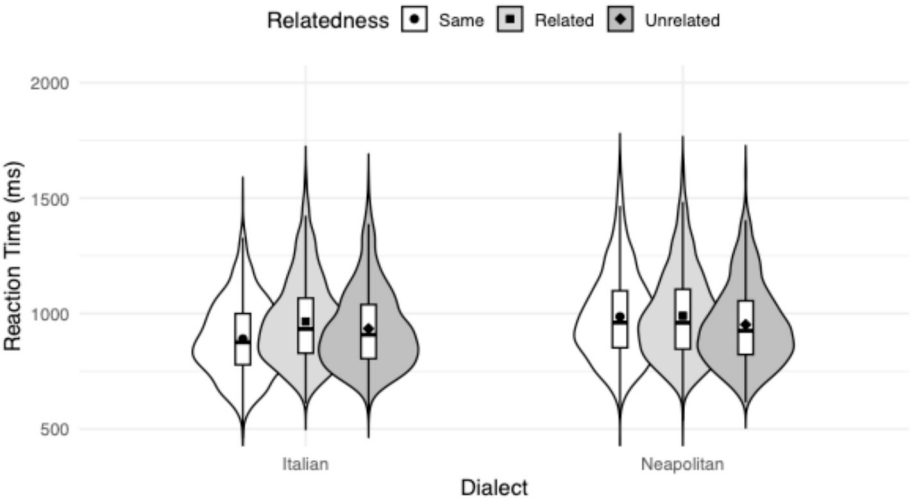


Fig. 2. Violin plots showing the distribution of Reaction Times (RTs) across the three levels of Relatedness (Same, Related, Unrelated) for each Dialect condition (Italian and Neapolitan). Each violin represents the distribution of individual trial RTs. Embedded boxplots display the interquartile range and median, and the black points indicate mean RTs for each condition. Reaction times are truncated between 500 and 2000 ms for visualization.

Table 3
Fixed Effects Estimates.

Term	Estimate	SE	df	t	p
Intercept	956.933	19.320	34.597	49.530	< 0.001
Related (Same v Unrelated)	−15.7	5.8	23.8	−2.	=0.012
Related (Related v Unrelated)	23.6	3.5	23.9	7.3	< 0.001
Dialect (Italian v Neapolitan)	−23.5	1.6	8678.0	−14.6	< 0.001
Related (Same)* Dialect (Ital)	−25.9	2.3	8676.9	−11.4	< 0.001
Related (Related)* Dialect (Ital)	10.7	2.3	8679.5	4.7	< 0.001

Note. The reference condition for Related is the unrelated condition.

relationship between Italian and Neapolitan, much as has been suggested for English varieties.

The results mirror previous findings exploring the distractor effects from dialectal translations of the picture names. We argue here and elsewhere that these effects are informative as to the organization of a bi-dialectal lexicon because they contrast with findings from bilinguals performing a similar task. However, we have never directly tested whether translation equivalent facilitation effects can be detected using spoken distractors; all demonstrations of translation equivalent facilitation effects have been observed using written distractors. There is some cause to question whether spoken and written distractors would necessarily always show the same effects. For example, [Hantsch et al., \(2005\)](#) observed that semantic facilitation from category labels (picture = carp, distractor = fish) reversed to interference when the modality of the distractor was changed from written to spoken. [Melinger \(2021\)](#) investigated this issue by directly comparing dialect equivalent effects elicited from spoken and written distractors using British and American English varieties. She observed robust interference effects from dialect equivalent distractors with spoken distractors (Exp 1) and with written distractors (Exp 2), although the latter were much smaller than the former (28 ms of interference from distractors spoken with a British accent, 40 ms of interference from distractors spoken with an American accent compared to 16 ms of interference for written distractors). Hence, effects from auditory distractors have been directly compared to the more common method using written distractors and found to induce similar, albeit weaker, results.

On the other hand, observing interference effects with written distractors is not the same as demonstrating that auditory distractors can be used to detect translation equivalent facilitative effects. [Dylman and Barry \(2018\)](#) demonstrated that synonymous distractor words (e.g., target = sofa, distractor = couch) produced facilitation when participants named the less-preferred picture name, but no distractor effects were observed when they named the preferred picture label. In contrast, [Melinger \(2018, Exp 4\)](#) asked participants to produce the dispreferred dialectal picture label (e.g., target = breeks, distractor = trousers), rather than the standard picture label, and in that case no dialect equivalent interference was observed, in contrast to when participant produced the preferred label. While this contrasting pattern of results may reflect fundamental differences between synonyms and dialects, it may also reflect differences from the modality of the distractors. To apply the strongest test of the current methodology, in Experiment 2 we use the same design and modality of the distractors to test for translation equivalent facilitation from Italian – English bilinguals. Following on from [Melinger \(2021\)](#), we may not detect a facilitation effect, but anything short of an interference effect should suffice to discriminate between the two patterns of results.

3. Experiment 2: Italian-English bilinguals

3.1. Methods

3.1.1. Participants

Forty-one native Italians (37 included; 16 male, 19 female, 2 prefer not to say) who live in Italy and have studied English to a highly proficient level were recruited using the Prolific participant recruitment

service. Participants were compensated for their time at a rate of €10/h. Because the study was run online and the quality of the data could not be controlled during testing, we recruited a larger sample to allow for anticipated exclusions. In the end, the data quality was good and only four participants were subsequently excluded from the analysis. One participant was excluded because they completed the study in English rather than Italian. One was excluded because their recorded responses were too noisy to extract RTs and another was excluded because there was evidence in the recordings that they were eating while completing the study. One further participant was excluded due to not meeting the inclusion requirements of having high English proficiency (self-reported familiarity with only 6 of the 32 English stimuli). This left 37 participants to be included. Participants ranged in age from 20 to 60 with a mean age of 31.4. On average, they had been studying English for 13.3 years. The shortest period spent studying English was 5 years. The experimental procedure was approved by the University of Dundee Psychology Ethics Committee and participant gave written informed consent prior to testing.

Following the same procedure used for Experiment 1, the sensitivity analysis showed we could achieve 87 % power for an effect size (d) of 0.35 with 37 subjects and 32 items, each presented twice in every condition ([Westfall et al., 2014](#)). For paired comparison (e.g., to detect translation equivalent facilitation or interference) we could achieve 80 % power to detect an effect size of 0.48.

3.1.2. Materials

Of the 32 stimuli developed for Experiment 1, 15 needed to be replaced due to phonological similarity between the Italian and the English word. For example, the Italian and English words for *broccoli* are nearly identical and the Italian and English words for *carrot* / *carota* are cognate. Therefore, a new set of thirty-two common objects were selected, including 17 items originally from Experiment 1. The stimuli were drawn from 7 semantic categories and photographs for each object were selected. All objects had phonologically distinct Italian and English labels. An Italian consultant oversaw materials selection.

Each picture was paired with six auditorily presented distractor words. Distractor words were either Italian words that corresponded to the picture labels (e.g., Cipolla ‘onion’) or their English translation equivalents. Distractor words either referred to the same meaning as the target picture (e.g., target = CIPOLLA, distractors = cipolla OR onion), a semantically related meaning (e.g., target = CIPOLLA, distractors = porro OR leek), or an unrelated meaning (e.g., target = CIPOLLA, distractors = strega or witch). Distractor words were recorded by a native Italian female who had lived in the Pacific-Northwest of the United States and had a native-like, fairly neutral, American accent when speaking English. Stimuli were presented in 2 blocks consisting of 192 trials each, with the order of trials in each block varying across participants. In all other respects, picture and distractor word preparation was identical to Experiment 1. A full set of stimuli can be found in Appendix B.

As in Experiment 1, following the main body of the experiment, participants completed a language usage questionnaire, presented in Italian, and an English word knowledge test, assessing familiarity with the English vocabulary used in the experiment.

3.1.3. Procedure

In contrast to Experiments 1, this experiment was conducted online, using the Gorilla experiment presentation platform (<https://gorilla.sc/>). Because the testing setting could not be controlled by the experimenter, participants were instructed at the start of the study that they would need one hour of uninterrupted time to complete the experiment and that they should find a quiet location where they would not be disturbed or distracted.

Participants were familiarized with the stimuli and their Italian names. Pictures and their Italian names appeared one at a time on the computer screen and the participants advanced to the next item by pressing the space bar. Following this familiarization phase, participants practiced naming the pictures each once before the main PWI task commenced. However, because no experimenter was present during the testing session, there was no opportunity to correct naming errors. Responses during the practice block were recorded and checked for accuracy after the fact and any picture that was not named with the correct label were subsequently excluded from the analysis, see below for details. To ensure that adequate testing conditions were maintained through the study, participants were asked to confirm whether they were still in a quiet location without distractions between the practice block and the main experiment and again between the 2 experimental blocks.

Individual responses were recorded by the presentation software. These sound files were subsequently downloaded and CheckVocal (Protopapas, 2007) was used to extract RTs and check naming accuracy. Because the recording began at the start of the trial rather than with picture presentation, the extracted RTs were corrected by subtracting 350 ms from every RT (this accounts for the 200 ms of silence at the start of trials and the SOA of -150 ms). In all other respects, the procedure was identical to that of Experiment 1. A total of 384 trials were recorded for each participant, with a total of 14,199 trials in total.

3.2. Results

Table 4 reports the average rating for the frequency of use of each variant as well as the self-assessed proficiency in listening and speaking to both Italian and English. Paired *t*-tests reveal that this sample report using Italian more often than English, $t(36) = 10.3$, $p < .001$, and their self-assessed proficiency in Italian was also higher than for English, $t(36) = 9.2$, $p < .001$ for listening and $t(36) = 13.2$, $p < .001$ for speaking. On average, participants reported being familiar with 90.5 % of the English words used in the experiment.

Mean response times (RTs) for correct trials, standard deviations, and mean proportion of errors in the experimental conditions are presented in Table 5 and visualized in Fig. 3. Trials in which the participant produced the wrong word, stuttered, hesitated, or failed to respond, and trials with voice key failures were classified as errors and discarded from the RT analysis ($N = 654$; 4.6 %). Additionally, any trial which was 2.5 standard deviations from a participant's condition mean was treated as an outlier ($N = 324$; 2.5 %). In total, 13203 trials were included in the final analysis. As error rates were low (5 %), they were not analysed further.

Linear mixed effects models were used to analyse the reaction times

Table 4

Mean (SD) rating estimates for the frequency of use of each variant, the self-reported proficiency in each variant, and the mean number of experimental dialect words that participants identified as familiar.

	Italian	English
Frequency of use	5.0 (0.2)	3.2 (1.0)
Listening Proficiency	5.0 (0)	3.7 (0.8)
Speaking Proficiency	5.0 (0)	3.1 (0.9)
English word familiarity		90.5 % (0.1)

Table 5

Mean response times (RTs) for correct trials, standard deviation and mean proportion of errors in the experimental conditions.

Distractor Conditions	Language			
	Italian		English	
	RT (errs)	SD	RT (errs)	SD
Same meaning	1050 (0.02)	186	1161 (0.06)	186
Same category	1190 (0.04)	194	1177 (0.06)	191
Unrelated	1162 (0.04)	190	1149 (0.05)	182
Identity effect (unrelated – same meaning)	+112 (0.02)		–11 (–0.01)	
Categorical Interference (unrelated – same category)	–28 (0.00)		–28 (–0.01)	

across conditions, following the same procedure outlined for Experiment 1. The model included *Language* (Italian, English) and *Relatedness* (same meaning, related meaning, unrelated) as categorical fixed factors. Participants and items were included as random effects. The final model included random slopes of *Relatedness* by participants.² Follow-up comparisons on the model's estimated marginal means used the Satterthwaite approximation for degrees of freedom and the Tukey correction for multiple comparisons.

The results of this omnibus analysis, presented in Table 6, revealed a significant main effect of *Relatedness* that is driven by overall faster naming times in the same meaning conditions compared to the unrelated conditions and slower naming times in the same category conditions compared to the unrelated conditions. Estimated Marginal Means (averaged across *Language*) confirmed that participants named pictures faster in the same meaning condition compared to the unrelated condition, $M_{diff} = 50$ ms, $z = -7.6$, $SE = 6.7$, $p < .001$, but slower in the related meaning condition compared to the unrelated condition, $M_{diff} = 28$ ms, $SE = 4.9$, $p < .001$.

There was also a significant main effect of *Language*; naming times were slower when the distractor was in English compared to Italian, when averaged across levels of *Related*, $M_{diff} = 28.7$, $z = 8.9$, $SE = 3.22$, $p < .001$.

The interactions between *Relatedness* and *Language* were significant for both the same meaning versus unrelated comparison and the related meaning versus unrelated comparison, as seen in Table 6. To clarify these interactions, two additional LMEs were conducted with a reduced fixed effects structure. When the related meaning condition was removed from the model, the interaction between *Relatedness* (same vs. unrelated) and *Language* remained significant, $t(8745.2) = 15.8$, $B = -30.9$, $SE = 2.0$, $p < .001$. In contrast, when the same meaning condition was removed, the interaction between *Relatedness* (related vs. unrelated) and *Language* was not significant, $t(8666) = 0.145$, $B = -0.3$, $SE = 2.0$, $p = .885$. This reflects the fact that the size of the semantic interference effect was comparable across languages whereas the impact of same-meaning distractors was modulated by *Language*, as seen in Table 5.

Planned comparisons, investigated the nature of the *Language* x *Relatedness* interactions. Participants named pictures more quickly when the Italian distractor word conveyed the same meaning as the target picture, relative to the Italian unrelated condition, $M_{diff} = -112$ ms, $z = 14.5$, $SE = 7.7$, $CI [-127.4, -97.1]$, $p < .001$, while the English same meaning distractors had no reliable impact on naming time relative to the English unrelated condition, $M_{diff} = 11$ ms, $z = 1.5$, $SE = 7.7$, $CI [-4.0, 26.5]$, $p = .47$. Additionally, both Italian and English related distractors slowed picture naming times relative to their language-specific controls. Traditional semantic interference effects were observed from Italian distractors, $M_{diff} = 27.7$ ms, $z = 4.4$, $SE = 6.3$, CI

² $RT \sim Related * Language + (1 + Related | subject) + (1 | picture)$

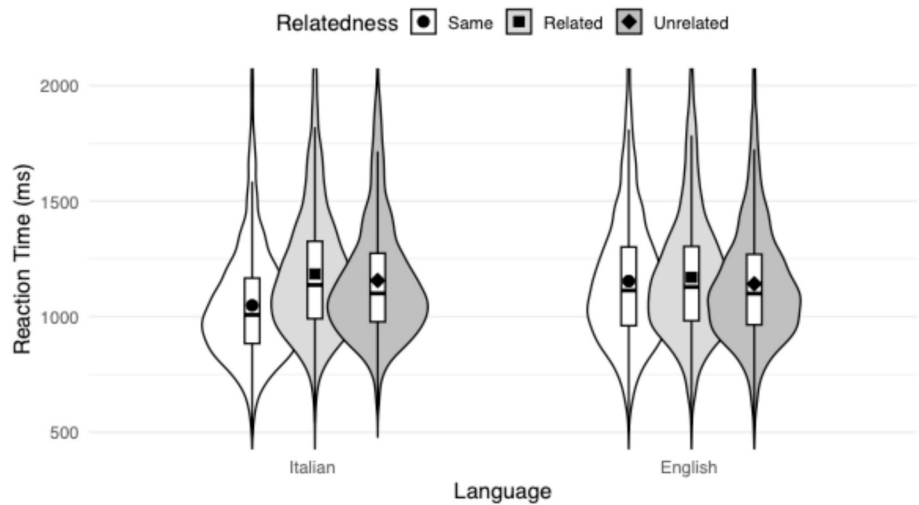


Fig. 3. Violin plots showing the distribution of *Reaction Times (RTs)* across the three levels of *Relatedness* (*Same, Related, Unrelated*) for each *Language* condition (*Italian and English*). Each violin represents the distribution of individual trial RTs. Embedded boxplots display the interquartile range and median, and the black points indicate mean RTs for each condition. Reaction times are truncated between 500 and 2000 ms for visualization.

Table 6
Fixed Effects Estimates

Term	Estimate	SE	df	t	p
Intercept	1149.7	31.9	43.0	36.1	< 0.001
related (Same v Unrelated)	−43.0	4.4	35.7	−9.6	< 0.001
related (Related vs Unrelated)	35.6	3.6	34.7	9.8	< 0.001
language (Italian vs English)	−14.3	1.6	13,059.8	−8.9	< 0.001
related (Same) * language	−41.0	2.3	13,061.5	−18.0	< 0.001
related (Related) * language	20.2	2.3	13,061.7	8.8	< 0.001

Note. The reference condition for Related is the unrelated condition.

[15.3, 40.0], $p < .001$, and from English distractors, $M_{diff} = 28.7$ ms, $z = 4.5$, $SE = 6.3$, $CI [16.3, 41.0]$, $p < .001$. While the apparent 11 ms difference between the English same meaning distractor condition and the English unrelated distractor condition might suggests a trend towards interference, it must be noted that this mean is inflated by a single outlier participant who produced a 237 ms interference effect, which is more than 2.5SD from the mean effect in this condition. If this single subject's data is removed, the comparison reduces, $M_{diff} = 5$ ms, $z = 0.7$, $SE = 7.2$, $CI [-9, 19.2]$, $p = .5$, while all other effects remain unchanged.

3.3. Discussion

Experiment 2 tested for between-language effects from auditory distractor words. Robust identity and semantic interference effects were observed for both Italian and English distractors. Crucially, we were interested in the polarity of the between-language translation equivalent distractor effect. Here, the results were not straightforward. English translation equivalent distractors induced no reliable influence on naming times. Prior studies using written distractors observe facilitation in this condition. The aim of this study was to ensure that the interference observed for translation equivalent dialect distractors was not due to the modality of the distractor presentation. [Hantsch, Jescheniak, and Schriefers \(2009\)](#) found a polarity reversal for level of specificity effects when the modality of the distractor was changed, with written distractors revealing facilitation but auditory distractors producing interference for the same distractor relationships. [Melinger \(2021\)](#) tested this possibility by eliciting dialectal equivalent effects from both auditory and visual distractors. She demonstrated that dialect equivalent interference can be observed with written distractors, hence arguing that distractor modality was not a critical factor in these experiments.

However, the robustness and magnitude of the interference effect observed with written distractors was reduced relative to auditory distractors, as mentioned above. Melinger did not conduct a between-experiment analysis to probe differences arising from the modality of distractor presentation, limiting our ability to now interpret this numerical pattern. However, the pattern is consistent with the claim that visual distractors bias towards facilitation in some instances relative to auditory distractors ([Hantsch et al., 2009](#)). The role of distractor modality will be further explored in the General Discussion.

To assess whether the results from Experiment 1 and 2 were qualitatively different, a direct comparison of the translation equivalent effects is required. To this end, we conducted a between experiment LME including 2 levels of the factor *Relatedness* (translation equivalents vs unrelated) and *Variant* (Neapolitan (Exp 1) vs. English (Exp 2)) as fixed factors. Subjects and pictures were included as random factors, noting that some pictures were common across experiments, while others were not. The final model included random slopes for *Relatedness* and *Variant* and their interaction by Subjects and Relatedness by Items.³ This analysis revealed significant main effects of Relatedness, $t(60) = 2.3$, $SE = 5.3$, $p = .025$, and Variant, $t(61) = 4.6$, $SE = 19.7$, $p < .001$. Critically, it also revealed a significant interaction between Relatedness and Variant, $t(97) = 2.24$, $SE = 3.55$, $p = .028$. This interaction reflects the fact that the interference observed from translation equivalent Neapolitan distractor words, $M_{diff} = 40$ ms, $z = 3$, $SE = 13.3$, $CI [14, 66]$, $p = .003$, was qualitatively different to the patterns of naming times when Italian speakers are presented with translation equivalent distractor word from English, $M_{diff} = 8.6$, $z = 0.71$, $SE = 12.2$, $CI [-15, 32]$, $p = .48$, supporting the conclusion that Neapolitan is represented in a co-

³ $RT \sim \text{Related} * \text{Variant} + (1 + \text{Related} * \text{Variant} | \text{subject}) + (1 + \text{Related} | \text{picture})$

dependent fashion while English is not.

4. General Discussion

The aim of this study was to examine the contribution of prestige, usage rates, and mutual intelligibility to the establishment of language-like lexical representations and organization. To this end, in Experiment 1, we tested for translation equivalent distractor effects from Italian native speakers who were also proficient in Neapolitan. We used the polarity of translation equivalent distractor effects to diagnose whether Neapolitan is represented and processed as a dialect of Italian or as an independent language. Consistent with previous studies looking at lexical selection across dialect boundaries, we observed that Neapolitan translation equivalents competed for selection with the Italian target word, supporting the view that Neapolitan is a dialect. These findings replicate previous results obtained in more formal laboratory settings (Melinger, 2018, 2021) with participants who were, in some instances, passive speakers of the dialect. The important revelation from this experiment is that manipulating linguistic properties such as proficiency and mutual intelligibility, as well as changing the testing environment, does not impact the polarity of the translation equivalent distractor effect. Critically, prior studies using the same design but testing “recognised” languages in a bilingual population have repeatedly reported facilitation effects when visually presented translation equivalent distractor words are paired with a target picture for a naming response. If proficiency or mutual intelligibility were a critical factor in determining the polarity or magnitude of these effects, we would have expected to find reduced interference effects for Neapolitan compared to the results previously obtained for English varieties. We saw no evidence for this.

In Experiment 2, we attempted to replicate the translation equivalent facilitation effect with auditory distractors. While we did not observe facilitation, we also did not observe reliable interference. Similar to the pattern that emerged in Melinger (2021) when examining synonym facilitation, a null effect was observed. A post-hoc between-experiment analysis revealed that the interference effect from Experiment 1 was reliably different to the null effect observed in Experiment 2, further supporting the conclusion that these variants are represented and processed differently. This result will be discussed further below in the context of a selection-by-competition word production model.

The only other instance where facilitation was not observed for translation equivalent distractors comes from Geukes and Zwitterlood (2016). They trained German native speakers with newly learned artificially generated picture names. After training, they conducted a similar PWI experiment testing for semantic interference and translation equivalent effects. They observed a translation interference effect when naming pictures in their L1, which they interpreted as evidence that newly learned words are not associated with a language tag, hence they cannot produce the language-based inhibition ostensibly required to observe a facilitation effect. This argument suggests that proficiency and usage underpin the establishment of a language tag, which seems self-evident. The present results go further to suggest proficiency and usage may be necessary, but not sufficient, conditions.

The broader pattern of results across the present experiments is consistent with theoretical predictions and the empirical record. When the distractor word expressed a categorically-related concept, either in the target variant or the alternative variant, target picture naming times were slower relative to their variety-specific unrelated control. The magnitudes of these inference effects were similar across variants, across experiments, and relative to the translation equivalent interference effect. This latter observation is consistent with results from the PWI literature suggesting that semantic distance does not impact on the magnitude of semantic interference effects unless the study is specifically designed to test for such effects (e.g., Mahon, Costa, Peterson, Vargas, & Caramazza, 2007; Aristei & Abdel Rahman, 2013, but see Rose, Aristei, Melinger, & Abdel Rahman, 2019). Similarly, when the distractor word expresses the same meaning as the target picture in the

same variety as the response, target naming times are facilitated, as would be expected in an identity condition.

4.1. Dialect usage & proficiency

For the first time, an investigation of dialect translation equivalent effects considered the proficiency and dialect usage patterns of the participants taking part. Prior work (e.g., Dal Negro & Vietti, 2011; Smith & Durham, 2012) has noted that dialectal usage rates and proficiency can vary wildly within a population, especially across generations and socio-economic bands. Evidence also has suggested that active vs. passive dialect usage can impact the processes used by speakers and listeners (Sumner & Samuel, 2009). For example, Sumner and Samuel investigated the input representations for speakers of a New York City variant that has a reduced rhotic word final production (e.g., baker → bak[ə]). They argued that overt dialect users, defined as those that produced the r-less forms in exit interviews, establish separate input representations for the dialectal (reduced) and standard (unreduced) word forms, while the covert dialect users, defined as those who did not actively produce the r-less forms, and those unfamiliar with the dialect utilized a shared, common, input representation when processing both the full and reduced forms.

In the present investigation, testing speakers who are active and proficient speakers of their respective regional variants still resulted in a dialect equivalent interference effect. Although no direct comparison with other groups was conducted, the size of the interference effect was comparable to previously reported effects (e.g., 27 ms interference in Experiment 1 of Melinger, 2018 compared with 35 ms interference in the present Experiment 1). We hypothesized that, if active experience speaking and listening to a dialect supports the establishment of independent linguistic systems, then the Italian-Neapolitan sample might have exhibited distractor effects more akin to those observed in bilinguals, resulting in translation equivalent facilitation or at least an attenuated interference effect. However, this prediction was not supported. Active use and increased proficiency does not seem to be the necessary and sufficient criterion for language-like linguistic independence.

It should be noted that the present study relied on self-reported proficiency ratings. We asked participants to rate their competency in speaking and listening, to assess their frequency of use of their different linguistic varieties, as well as complete a questionnaire assessing their familiarity with the vocabulary used in the experiment. This approach to gauging proficiency combines commonly used approaches from the literature, noting that across studies of translation equivalent distractor effects there is no standard approach. In fact, some studies do not assess proficiency at all, instead relying on sociolinguistic reports of the distribution of language usage within a population (e.g., Feleke, 2024). However, most studies do report self-rated competency across the four performance domains of reading, writing, listening, and speaking. Many additionally report variables such as the age at which the 2nd language education began or an estimate of the time spent using each language (e.g., Costa & Caramazza, 1999; Hermans, 2004).

However, recent research by Tomoschuk et al. (2019) has raised questions about the validity of self-reported proficiency ratings, observing within and between group differences in how the rating scale is interpreted. They note that factors such as language dominance, learning environment, and language combination influence how an individual interprets the points on a self-rated Likert scale. As a result, they caution against using self-reported proficiency for between-group comparisons and support the increased use of objective proficiency ratings, such as the Multilingual Naming test or MINT (Gollan, Weissberger, Runnqvist, Montoya, & Cera, 2012) or proficiency interviews. Some recent translation equivalency investigations have included objective measures of proficiency (Emmorey et al., 2021; Tomoschuk et al., 2021).

The findings from Tomoschuk et al. (2019) suggest extreme caution

should be exercised when comparing proficiency levels across groups of different language / dialect pairs. Here, we have argued that the bidialectal speakers were highly proficient. This claim was supported by environmental factors (e.g., patterns of dialect usage reported for the geographical region and/or associated with the socio-economic status of the region) as well as by self-reported familiarity with the vocabulary used in the experiment, self-reported usage rates for the two linguistic varieties as well as self-reported proficiency ratings. Therefore, this essential claim is supported by more than just the self-rated proficiency ratings.

4.2. Mutual intelligibility

The varieties of English that have been assessed to date are broadly mutually intelligible, particularly after a period of perceptual learning and adaptation (e.g., Bradlow & Bent, 2008; Eisner, Melinger, & Weber, 2013). Mutual intelligibility and formal similarity may enable the sharing of lexical and grammatical representations, reducing the need for separate or independent stored representations for the two varieties. To illustrate, consider the English word 'walking'. English speakers can vary the formality of their speech by truncating the suffix, producing *walkin'* instead of the full form *walking*. But the co-existence of these two forms does not necessitate the postulation two lexical representations for the formal and informal variants because the informal can be regularly derived from the full form. Perhaps an analogy can be drawn between register variation and regional dialectal forms with high rates of form similarity. If two regional varieties are sufficiently similar as to be mutually intelligible, with large proportions of their lexical and grammatical inventory being distinguished by regular or rule-governed alternations, then perhaps they too would not need distinct lexical and grammatical representations. In such a case, only the idiosyncratic differences would need to be stored and tagged as belonging to that specific dialect, producing a smaller, dependent sub-section of the lexicon and grammar for uniquely dialectal forms. Such a situation is consistent with the co-dependent lexical organization proposed by Labov, 1998 and embraced by Melinger (2018, 2021; see also Heggdal Lønes et al., 2023) to explain the contrasting effects of translation equivalent distractors.

However, the Neapolitan variety examined in Experiment 1 captures a very different scenario. Neapolitan and Standard Italian are reported as being mutually non-intelligible (ELAlliance, n.d.), despite of the relative closeness as computed by form similarity measures (Boga, 2020). This presents an opportunity to test the hypothesis outlined above. A lack of mutual intelligibility may set the stage for the development of distinct lexical and grammatical representations, which in turn results in independence between varieties. Note, this contrast between co-dependent and independent organization is hypothesized to underpin the psychological distinction between dialects and languages. Experiment 1 was intended to test this hypothesis and the results did not support the proposal. Despite the low levels of mutual intelligibility, Italian - Neapolitan speakers still produced interference when naming pictures while hearing distractors that conveyed the same meaning as the picture label in the non-target variety. There are two ways to interpret this result. Either mutual intelligibility, even combined with active usage and high proficiency, is not a necessary and sufficient condition for the establishment of independent, language like, representations or the variants chosen were not sufficiently different to satisfy this criterion. It is unclear, at this point, which of these interpretations is correct.

4.3. Testing context

In the original investigations reported by Melinger (2018), SSE-Scot bidialectal participants were tested in a formal, university laboratory setting. Participants were drawn from the university undergraduate population and many participants self-reported that they rarely used Scots dialects, expressing a view that the repertoire represented 'bad

English' or slang. Given the association between dialect usage and informal contexts, we speculated that experimental testing in a formal setting, such as a psychology laboratory on a university campus, might have inhibited participants during testing, producing artificially-inhibited responses. In the present study, all participants were tested in their home environments, either in person (Exp 1) or online (Exp 2). The change in context did impact overall RT patterns, as mean RTs in Experiment 1 were approximately 100 ms slower than in similar laboratory-based experiments (Melinger, 2018, 2021) and the mean RTs in Experiment 2 were slower again by a further 200 ms. Standard deviations in Experiment 1 were comparable to previous studies, averaging around 85 ms yet the SDs in Experiment 2, when no experimenter was present, doubled to around 180 ms.

Yet despite these changes to the testing context and the slower naming times, the pattern of results was broadly consistent with previous findings obtained in the formal university setting. Semantic interference, both within and between varieties, was observed as was a robust identity effect in both experiments. Error rates were also within normal parameters.

This is not the first study to report slower RTs for data collected outside the laboratory. Fairs and Strijkers (2021) compared online picture naming times to those collected in the laboratory and found a similar naming time slowdown, with RTs more than 200 ms slower than the laboratory RTs (means ~1100 ms). Vogt et al., (2022) similarly found slower RTs, exceeding 1000 ms, when using their preferred JsPsych platform compared to the in-person data, which averaged around 875 ms in comparison. Despite these slower naming times, both studies successfully replicated in person experiments. Fairs & Strijkers found word frequency effects of a similar magnitude to those observed from faster RTs collected in the laboratory context and Vogt et al. replicated the traditional semantic interference effect in a PWI paradigm. Fairs & Strijkers speculated that the absence of an experimenter might have removed the time pressure on participants to respond. A similar explanation might be at play in the current studies. Similarly, testing people in their own homes, as was done in Experiment 1, might create a more relaxed environment compared to the laboratory, leading to the slower naming times observed in Experiment 1, even though an experimenter was still present in that case. Overall, what is important to note is that the emergence of well-established effects, even when RTs are slower, should give confidence that the distractor effects tested in the current study are robust to changes in testing context.

4.4. The role of distractor modality

In Experiment 2, we failed to replicate the well-known translation equivalent facilitation effect when distractor words were presented auditorily. This is a critically important observation, as many linguistic varieties do not have a standardised written form, so auditory distractor presentation may be the only option available to investigate translation equivalent processing. Previously, Melinger (2021) argued that distractor modality did not impinge on translation equivalent distractor effects. However, the results of Experiment 2 challenge this previous claim. What mechanism might account for modality effects?

Hantsch et al. (2009) examined the role of *response congruency* in basic and superordinate picture naming. Response congruency is when, given the task instructions, the target and the distractor would both produce the same response. For an observable effect of response congruency to emerge, representational convergence is needed at both the conceptual and lexical levels. For example, imagine a situation where a picture of a poodle is presented with a distractor 'labrador' and the task is to name the picture at the basic level, i.e., to say 'dog'. In this case, following the task instructions, both stimuli converge on the same response, speeding processing. Hantsch et al. observed that, when the distractors were presented auditorily, this expected facilitation effect flipped to interference. The explanation offered was that presenting both target and distractor in the visual modality enhanced the conceptual

level convergence effect, strengthening the conceptual facilitation effect relative to any potential lexical competition effect. In contrast, when the target and distractor are presented in different modalities, no strengthening of the conceptual effects occurred, allowing lexical competition to win out.

This explanation, which refers implicitly to the tradeoff between conceptual and lexical processes, is reminiscent of the tradeoff account fundamental to the Swinging Lexical Network account of semantic context effects (Abdel Rahman & Melinger, 2009, 2019). This proposal is outlined below and provides a framework for interpreting the present results.

4.5. Models of bidialectal lexical organization

The key observation from investigations of dialect equivalent distractors is that they interfere with picture naming while between-language translation equivalent distractors facilitate naming. Melinger (2018) argued that this contrast supported a qualitative representational distinction between independent and co-dependent lexical organizations. However, facilitation from synonymous distractors (Dylman & Barry, 2018) challenges this account. Hence, the initial assertion that translation equivalent interference was a hallmark of the monolingual lexicon needs to be refined. Instead, it seems that dialect equivalent interference is a hallmark of a sociopragmatic distinction in the lexicon. Following (Melinger, 2021; see also Heggdal Lønes, Kamide and Melinger, 2023, here I situate translation equivalent effects within the Swinging Lexical Network (SLN) model of word production (Abdel Rahman & Melinger, 2009, 2019).

Prominent models of bilingual lexical organization (De Bot, 1992; De Bot & Schreuder, 1993; Green, 1986, 1993, 1998; Poulisse & Bongaerts, 1994) propose that languages maintain their independence and separation via language membership tags, which sit outside the conceptual network and can inhibit representations from the non-target variant, as depicted on the left panel of Fig. 4. Co-dependence, in contrast, refers to sociolinguistically-marked lexical alternatives forming part of a single lexicon, as depicted in the right panel of Fig. 4. By including sociolinguistic and sociopragmatic features within the conceptual network, these representations can capture contextual and pragmatic meaning distinctions (Melinger, 2018, 2021, see Blanco-Elorrieta & Caramazza, 2021; La Heij, 2005 for similar proposals).

The SLN was designed to explain semantic context effects from a range of experimental paradigms and from different types of semantic relations (categorical, thematic, ad hoc, etc.). It has two key assumptions. First, it assumes lexical selection times reflect the net effects of semantic and lexical processing. Distractor effects at the semantic level are facilitative, as no decision is needed at this stage. Activation from the

target and distractor either converge on a common representation or diverge onto related representations. Distractor effects at the lexical level are interfering, as the selection mechanism requires one candidate to out-compete all other representations to be selected. Hence, the more strongly candidates are active, the longer selection time will take. These opposing effects trade off. If semantic facilitation outweighs lexical interference, facilitation will be observed, but both effects are always present. The second key assumption of the SLN is that competition effects take place over the entire lexicon, not only between the target and distractor. It is only when a strong cohort of lexical alternatives are activated that lexical interference will outweigh semantic facilitation.

This second assumption is key to explaining why synonymous distractors produce facilitation when dialectal equivalents produce interference. If the conceptual representations of the dialectal variants are not isomorphic, by virtue of the additional dialectal feature seen in the right panel of Fig. 4, the activation from the picture and distractor word will spread across the network to related concepts, much like what happens for category sisters. This activation spread results in a cohort of active lexical candidates that compete for selection. This many-to-one competition underpins observable interference effects. In contrast, if synonyms or translation equivalents share a conceptual representation, then the activation from the picture and distractor will converge on that shared representation, producing strong semantic priming but not strongly activating much else. This activation will spread to the two associated lexical candidates, producing one-to-one competition, which is not sufficient to offset the facilitation that was generated by the convergence of 2 sources of activation at the semantic level (see Abdel Rahman & Melinger, 2009, 2019 for an extended development of this argument and associated evidence base). Hence, the underpinning difference between synonyms and sociolinguistically-marked alternates is in the isomorphism of their semantic representations and the associated consequences for activation spread across the network.

This trade-off framework may offer any explanation for the modality reversals observed by Hantsch et al. (2009) as well as the absence of a detectable facilitation effect in Experiment 2. Hantsch et al. suggested that the common visual modality shared by picture and written distractor might enhance conceptual priming effects relative to cross-modal presentations of auditory distractors. In contrast, lexical processes should be unaffected by such perceptual features. As a result, when the distractor and picture converge on highly overlapping representations, as in the case for levels of specificity, for synonyms, and for translation and dialect equivalents, the differences in conceptual priming may be sufficient to tip the net effect from facilitation to a null effect or from a null effect to interference, as has been described for various effects within this literature above.

Note, this account does not require language tags to explain the

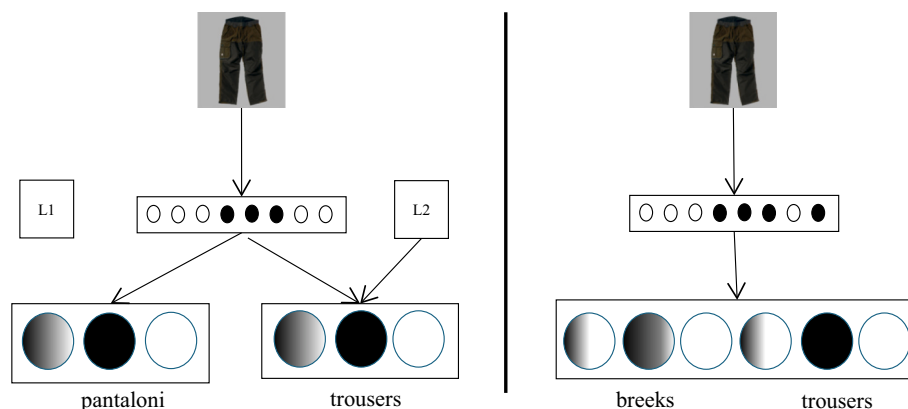


Fig. 4. Independent (left frame) and co-dependent (right frame) lexical organization (adapted from Kroll, Bobb, & Wodniecka, 2006). On the left panel, the target response is *pantaloni*, and the L2 language tag inhibits English alternatives. On the right panel, the target response is *trousers* – an additional dialectal feature at the conceptual level distinguishes the conceptual representations of dialectal alternates, leading to stronger activation of the appropriate lexical item.

polarity of the distractor effect, but language tags are needed to ensure the lexical alternative from the target language is ultimately selected. That is a key difference to synonyms and possibly for dialectal and register alternatives – the risks of miscommunications due to selection errors are low-to-nil for synonyms and sociolinguistic alternatives. In contrast, selecting a word from the unintended language will have significant communicative consequences. Hence this account still relies on a language control mechanism.

4.6. Summary

The question of how to distinguish between dialects and languages has plagued linguists for decades. The present investigation represents part of a wider effort to use behavioral observations obtained through well-established psycholinguistic paradigms to objectively distinguish between bilingual and bidialectal translation effects. Consistent with prior investigations of dialectal lexical selection studies, we observed translation equivalent interference between varieties that were mutually non-intelligible and in participants who were active, proficient speakers of both varieties, even when testing was conducted in a familiar environment. Based on the evidence presented, we can conclude that Neapolitan is represented and processed like a dialect of Italian rather than an independent language, much like varieties of English.

Here, we have taken another step in trying to pinpoint the necessary and sufficient characteristics required to distinguish between languages and dialects. While the critical features remain elusive, investigations such as this one promise to improve our understanding of how different types of linguistic varieties are represented and processed and should deliver an answer to the long-standing puzzle of how to objectively define a language.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT in

order to create Figs. 2 and 3 in GGPlot. After using this tool/service, the author reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Alissa Melinger: Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

None.

Data availability

All data, stimuli, and analysis files for the experiments reported in this paper are available on the OSF site here: osf.io/7zsuj/overview.

Acknowledgements

None of this work could have been achieved without the support and effort of my students. Many thanks go to Andrea Bonato for developing the stimuli for Experiment 1 and spending her Christmas holiday collecting and coding the data, and to Max McDonald and Mickey Singh for their heroic efforts developing stimuli, constructing the study on Gorilla, creating unique randomisations, and coding the data for Experiment 2. Additional thanks go to Sarah Allen for recording the distractors for Experiment 2 and to Tanja Atanasova for helping translate all the experimental documents into perfect Italian. Figs. 2 and 3 were created with assistance from ChatGPT. All errors and omissions are my own.

This work was supported through the School of Humanities, Social Sciences and Law Strategic Development and Impact Funding pathway at the University of Dundee.

Appendix A. Stimuli used in Experiment 1

Picture	Same meaning		Same category		Unrelated	
	Italian	Neapolitan	Italian	Neapolitan	Italian	Neapolitan
Carrot	Carota	Pastenacchia	Sedano	A'Laccia	Strega	Janara
Pig	Maiale	Titone	Gatto	Jat	Pantaloni	Cazoni
Witch	Strega	Janara	Macellaio	Chianchiere	Carota	Pastenacchia
Apron	Grembiule	Mantesino	Pantaloni	Cazoni	Arancia	Portuallo
Celery	Sedano	A'Laccia	Carota	Pastenacchia	Imbuto	Muto
Cat	Gatto	Jat	Maiale	Titone	Serratura	Mascatura
Butcher	Macellaio	Chianchiere	Strega	Janara	Sedano	A'Laccia
Trousers	Pantaloni	Cazoni	Grembiule	Mantesino	Foglia	Fronna
Pot	Pentola	Tiella	Vassoio	Guantiera	Ostetrica	Mammana
Lock	Serratura	Mascatura	Tetto	Pennata	Gatto	Jat
Ladle	Mestolo	Coppino	Forchetta	Brocca	Muratore	Fravecattore
Nettle	Ortica	Ardica	Broccoli	Friarielli	Lattina	Buatta
Tray	Vassoio	Guantiera	Pentola	Tiella	Ciliegia	Cerasa
Roof	Tetto	Pennata	Serratura	Mascatura	Falena	Palummella
Fork	Forchetta	Brocca	Mestolo	Coppino	Gabbia	Caiola
Broccoli	Broccoli	Friarielli	Ortica	Ardica	Pipistrello	Spurtiglione
Tin	Lattina	Buatta	Gabbia	Caiola	Maiale	Titone
Cherry	Ciliegia	Cerasa	Arancia	Portuallo	Tetto	Pennata
Bricklayer	Muratore	Fravecattore	Ostetrica	Mammana	Ghianda	Llanda
Leaf	Foglia	Fronna	Ghianda	Llanda	Pentola	Tiella
Cage	Gabbia	Caiola	Lattina	Buatta	Tacchino	Vallarine
Orange	Arancia	Portuallo	Ciliegia	Cerasa	Macellaio	Chianchiere
Midwife	Ostetrica	Mammana	Muratore	Fravecattore	Forchetta	Brocca
Acorn	Ghianda	Llanda	Foglia	Fronna	Caprone	Zimmero
Funnel	Imbuto	Muto	Bilancia	Valanzone	Asino	Sciecco
Moth	Falena	Palummella	Caprone	Zimmero	Mestolo	Coppino
Robin	Pettiroso	Reviezzolo	Tacchino	Vallarine	Grembiule	Mantesino

(continued on next page)

(continued)

Picture	Same meaning		Same category		Unrelated	
	Italian	Neapolitan	Italian	Neapolitan	Italian	Neapolitan
Donkey	Asino	Sciecco	Pipistrello	Spurtiglione	Vassoio	Guantiera
Scale	Bilancia	Valanzone	Imbuto	Muto	Pettiroso	Reviezzolo
Goat	Caprone	Zimmero	Falena	Palummella	Broccoli	Friarielli
Turkey	Tacchino	Vallarine	Pettiroso	Reviezzolo	Ortica	Ardica
Bat	Pipistrello	Spurtiglione	Asino	Sciecco	Bilancia	Valanzone

Appendix B. Stimuli used in Experiment 2

Picture	Same meaning		Same category		Unrelated	
	Italian	English	Italian	English	Italian	English
Onion	Cipolla	Onion	Porro	Leek	Strega	Witch
Leek	Porro	Leek	Cipolla	Onion	Imbuto	Funnel
Pig	Maiale	Pig	Cane	Dog	Camicia	Shirt
Dog	Cane	Dog	Maiale	Pig	Porta	Door
Ghost	Fantasma	Ghost	Strega	Witch	Porro	Leek
Witch	Strega	Witch	Fantasma	Ghost	Cipolla	Onion
Apron	Grembiule	Apron	Camicia	Shirt	Fragola	Strawberry
Shirt	Camicia	Shirt	Grembiule	Apron	Foglia	Leaf
Knife	Coltello	Knife	Vassoio	Tray	Ostetrica	Midwife
Tray	Vassoio	Tray	Coltello	Knife	Mela	Apple
Ladle	Mestolo	Ladle	Frusta	Whisk	Fantasma	Ghost
Whisk	Frusta	Whisk	Mestolo	Ladel	Gabbia	Cage
Roof	Tetto	Roof	Porta	Door	Cane	Dog
Door	Porta	Door	Tetto	Roof	Falena	Moth
Tomato	Pomodoro	Tomato	Aglio	Garlic	Acquario	Fishtank
Garlic	Aglio	Garlic	Pomodoro	Tomato	Pipistrello	Bat
Fishtank	Acquario	Fishtank	Gabbia	Cage	Maiale	Pig
Cage	Gabbia	Cage	Acquario	Fishtank	Pollo	Chicken
Apple	Mela	Apple	Fragola	Strawberry	Tetto	Roof
Strawberry	Fragola	Strawberry	Mela	Apple	Caprone	Goat
Hairdresser	Parrucchiere	Hairdresser	Ostetrica	Midwife	Ghianda	Acorn
Midwife	Ostetrica	Midwife	Parrucchiere	Hairdresser	Frusta	Whisk
Leaf	Foglia	Leaf	Ghianda	Acorn	Coltello	Knife
Acorn	Ghianda	Acorn	Foglia	Leaf	Pettiroso	Bat
Funnel	Imbuto	Funnel	Bilancia	Scale	Asino	Donkey
Scale	Bilancia	Scale	Imbuto	Funnel	Fantasma	Ghost
Moth	Falena	Moth	Pipistrello	Bat	Mestolo	Ladle
Bat	Pipistrello	Bat	Falena	Moth	Grembiule	Apron
Chicken	Pollo	Chicken	Struzzo	Ostrich	Aglio	Garlic
Ostrich	Struzzo	Ostrich	Pollo	Chicken	Pomodoro	Tomato
Goat	Caprone	Goat	Asino	Donkey	Vassoio	Tray
Donkey	Asino	Donkey	Caprone	Goat	Bilancia	Scale

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